TT-VLA Warm-up Fine-tuning Evaluation Report

Experiment Configuration

Base Model:	OpenVLA-7B
Fine-tuning:	LoRA (rank=32, alpha=32)
Training Data:	200 scripted demos (PickCube-v1)
Training Steps:	3,000 (early-stopped)
Final Train Loss:	0.216
Hardware:	8× NVIDIA RTX 5090 (DDP)
Training Speed:	3.3 steps/s
Control Mode:	pd_ee_delta_pose (7-DoF)

Evaluation Results (20 episodes, no TT-VLA)

Success Rate:	0.0% (0/20)
Avg Reward:	35.01
Std Reward:	23.80
Max Reward:	70.34
Min Reward:	5.89
Avg Steps:	160.0 (all hit max)
Avg Time/Episode:	51.4s

Key Findings

1. Non-zero rewards (avg 35.0) confirm warm-up learned meaningful manipulation behavior from demonstrations.

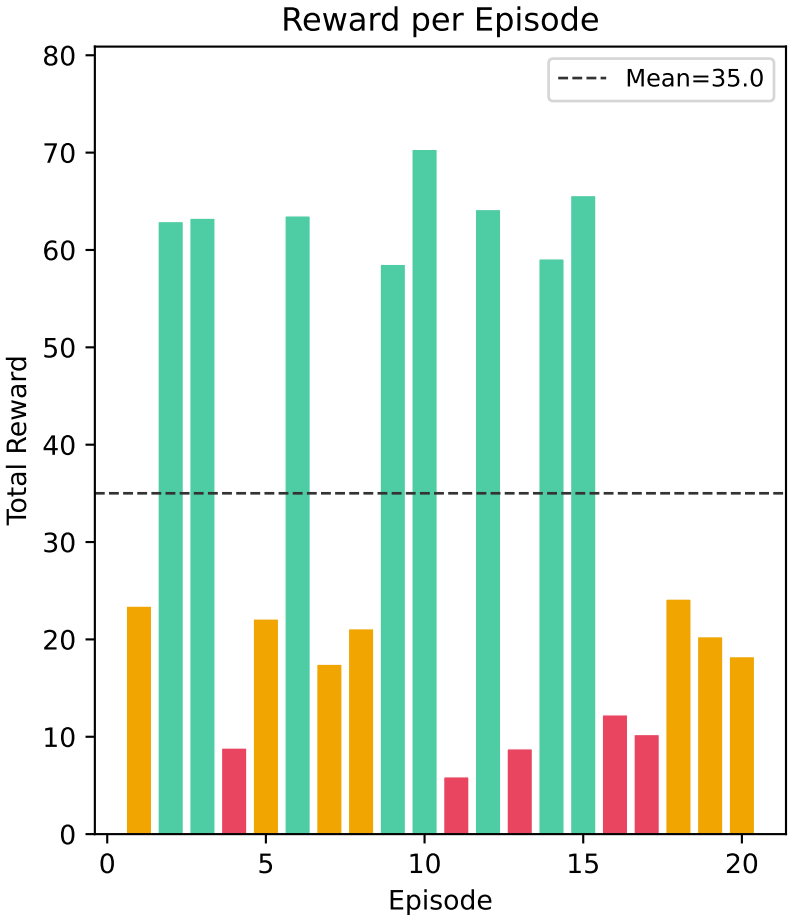
2. Bimodal distribution: 40% episodes reach high reward (>50), indicating partial task completion (approach + grasp attempt), while 25% show minimal progress (<15).

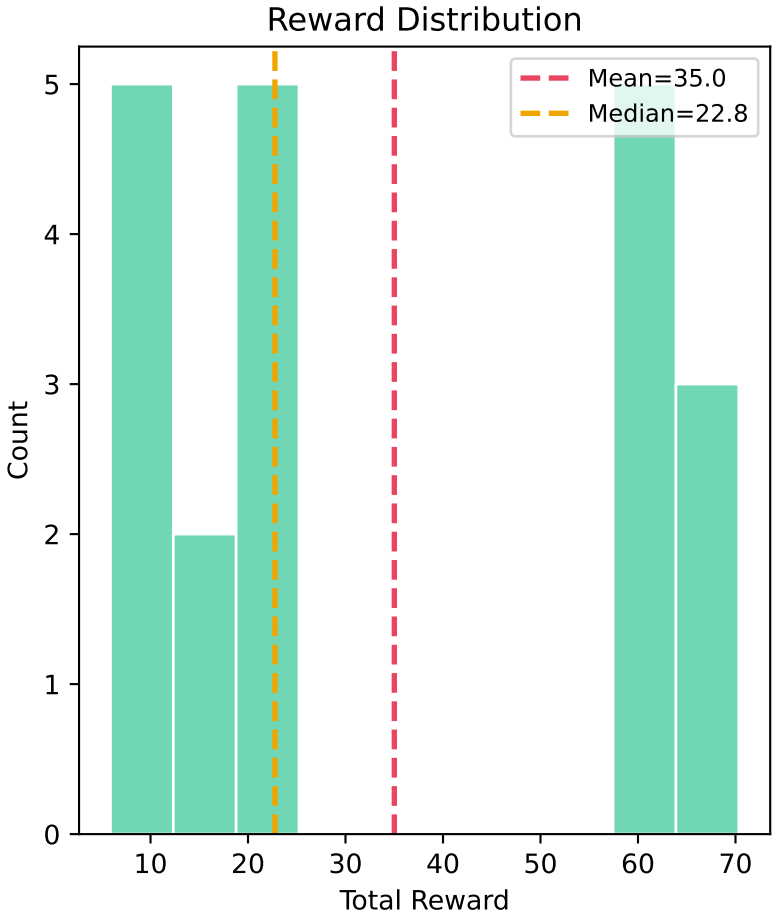
3. 0% success rate: full pick-and-place requires precise sequential execution that imitation alone cannot achieve.

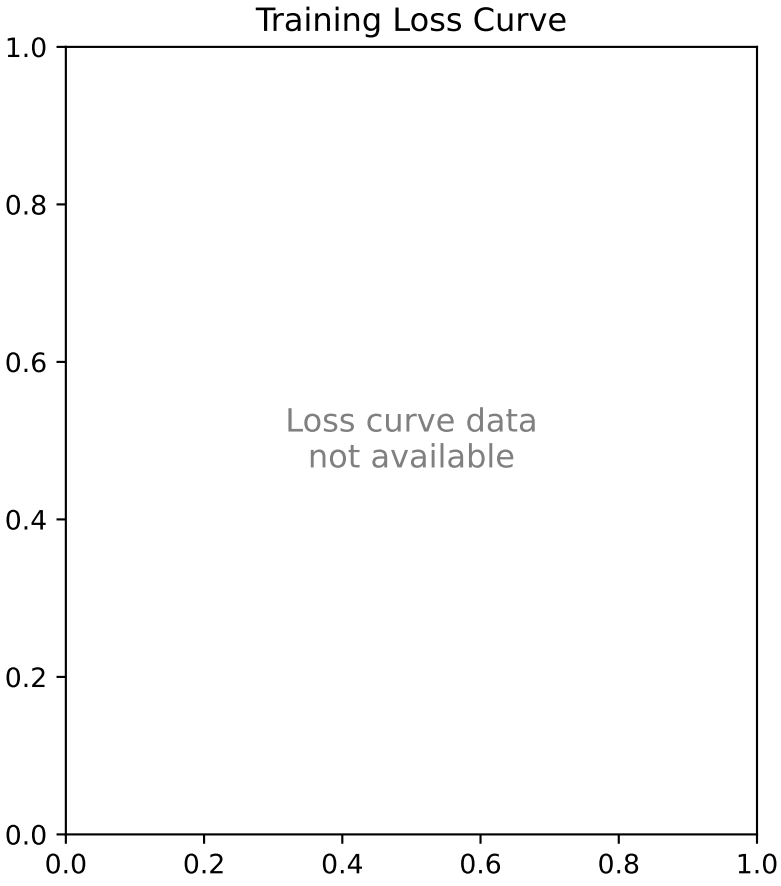
4. This gap motivates TT-VLA: test-time PPO adaptation should refine the warm-up policy to close the gap between partial progress and task completion.

Generated with TT-VLA | https://github.com/TT-VLA

Warm-up Evaluation: Detailed Results







Method Comparison

Method	SR	Avg Reward
OpenVLA (no FT)	~0%	~0
Warm-up only	0%	35.0
TT-VLA (target)	~45%	TBD